T3 C_-1: full general UV Lagrangian and RGEs

Matchete/RGBeta symbolic output in mathematical notation

C_-1

Additional quartics: 7, additional bilinears: 1, extra Yukawa: none.
General matching and Wilson coefficients are not computed in this symbolic report.

Notation key

symbol	interaction / definition
g_1	$D_\mu \supset i g_1 Y B_\mu$
g_2	$D_\mu \supset i g_2 T^A W_\mu^A$
g_3	$D_\mu \supset i g_3 t^A G_\mu^A$
Y_u	$-\bar{Q} Y_u \widetilde{H} u + \text{h.c.}$
Y_d	$-\bar{Q} Y_d H d + \text{h.c.}$
Y_e	$-\bar{L} Y_e H e + \text{h.c.}$
y_1	$L F S_1 + \text{h.c.}$
y_2	$L F S_2 + \text{h.c.}$
M_F	$-\bar{F} M_F F$
T	$\mathrm{Tr}(Y_e^\dagger Y_e + 3 Y_u^\dagger Y_u + 3 Y_d^\dagger Y_d)$
$T_\nu^{(1)}$	$\mathrm{Tr}(y_1^\dagger y_1)$
$T_\nu^{(2)}$	$\mathrm{Tr}(y_2^\dagger y_2)$
T_ν	$T_\nu^{(1)} + T_\nu^{(2)}$
m_1^2	$m_1^2 S_1^\dagger S_1$
m_2^2	$m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2}\lambda_1 (H^\dagger H)^2$
$\lambda_{S_1}^{(1)}$	$\frac{1}{2}\lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$
$\lambda_{S_2}^{(1)}$	$\frac{1}{2}\lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$
$\lambda_{HS_1}^{(1)}$	$\lambda_{HS_1}^{(1)} (H^\dagger H)(S_1^\dagger S_1)$
$\lambda_{HS_2}^{(1)}$	$\lambda_{HS_2}^{(1)} (H^\dagger H)(S_2^\dagger S_2)$
$\lambda_{HS_1}^{(A)}$	$\lambda_{HS_1}^{(A)} (H^\dagger T^A H)(S_1^\dagger T^A S_1)$
$\lambda_{HS_2}^{(A)}$	$\lambda_{HS_2}^{(A)} (H^\dagger T^A H)(S_2^\dagger T^A S_2)$
λ_5	$\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$
$\lambda_{12}^{(1)}$	$\lambda_{12}^{(1)} (S_1^\dagger S_1)(S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$\lambda_{12}^{(A)} (S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	$\lambda_{12}^{(\times)} (S_1^\dagger S_2)(S_2^\dagger S_1)$
$\lambda_{S_1}^{(A)}$	$\lambda_{S_1}^{(A)} (S_1^\dagger T^A S_1)(S_1^\dagger T^A S_1)$
$\lambda_{S_2}^{(A)}$	$\lambda_{S_2}^{(A)} (S_2^\dagger T^A S_2)(S_2^\dagger T^A S_2)$
α	$Y_{S_1} = \alpha/2,\ Y_F = (\alpha+1)/2,\ Y_{S_2} = (\alpha+2)/2$
$C_2(d)$	$(d^2-1)/4$
$T(d)$	$d(d^2-1)/12$
$C_{LLS_1S_2}^{ij;AB}$	$L_i L_j S_1^A S_2^B$
C_{ijab}	$L_i L_j \phi_a \phi_b$
C_5^{ij}	$(L_i^T C L_j) H H$
R_W	$16\pi^2 \beta_{C_5} \supset R_W \lambda_5 C_{LLS_1S_2}$

Complete BSM UV Lagrangian

Matchete BSM Lagrangian, excluding the pure Standard Model contribution. $\mathcal{L}_{\text{BSM}} = \text{Coupling(lambdaT3ExtraM000101I1, \{\}, 0)CG(T3ExtraCGM000101I1, \{Index(d\$1, T3BSMd2), Inde}$

$\text{+Coupling(lambdaT3ExtraQ0002I21, \{\}, 0)Bar(Field(NewScalar1, Scalar, \{Index(d881, T3BSMd2)\}, \{\}))Bar(Field(NewScalar1, Scalar, \{Index(d882, T3BSMd2)\}, \{\}))Bar(Field(NewScalar2, Scalar, \{Index(d883, T3BSMd2)\}, \{\}))Bar(Field(NewScalar2, Scalar, \{Index(d884, T3BSMd2)\}, \{\}))CG(T3ExtraCGQ0002I21, \{Index(d881, T3BSMd2), Index(d882, T3BSMd2), Index(d883, T3BSMd2), Index(d884, T3BSMd2)\})}$

$$+\text{Coupling}(\text{lambdaT3ExtraQ0202001}, \{\}, 0)\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2})\}, \{\}))\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd2})\}, \{\}))\text{Bar}(\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})\}, \{\}))\text{Bar}(\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})\}, \{\}))\text{CG}(\text{T3ExtraCGQQ0202001}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)}), \text{Index}(\text{d\$\$2}, \text{SU2L(fund)}), \text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Index}(\text{d\$\$2}, \text{T3BSMd2})\})$$

$$+\text{Coupling}(\text{lambdaT3ExtraQ110101I1}, \{\}, 0)\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2})\}, \{\}))\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd2})\}, \{\}))\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})\}, \{\})\text{Bar}(\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})\}, \{\}))\text{CG}(\text{T3ExtraCGQ110101I1}, \{\text{Bar}(\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})), \text{Index}(\text{d\$\$1}, \text{SU2L(fund)}), \text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Index}(\text{d\$\$2}, \text{T3BSMd2})\})$$

$$+\text{Coupling}(\text{lambdaT3ExtraQ110101I2}, \{\}, 0)\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2})\}, \{\}))\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd2})\}, \{\}))\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})\}, \{\})\text{Bar}(\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})\}, \{\}))\text{CG}(\text{T3ExtraCGQ110101I2}, \{\text{Bar}(\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})), \text{Index}(\text{d\$\$1}, \text{SU2L(fund)}), \text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Index}(\text{d\$\$2}, \text{T3BSMd2})\})$$

$$+\text{Bar}(\text{Coupling}(\text{lambdaT3ExtraQ0202001}, \{\}, 0)\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2})\}, \{\}))\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd2})\}, \{\}))\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})\}, \{\})\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})\}, \{\})\text{CG}(\text{Bar}(\text{T3ExtraCGQQ0202001}), \{\text{Bar}(\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})), \text{Bar}(\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})), \text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Index}(\text{d\$\$2}, \text{T3BSMd2})\})$$

$$+\text{Coupling}(\text{MS1}, \{\}, 0)^2\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2})\}, \{\})(-\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2})\}, \{\}))$$
$$+\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2})\}, \{\text{Index}(\text{d\$\$1}, \text{Lorentz})\})\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2})\}, \{\text{Index}(\text{d\$\$1}, \text{Lorentz})\}))$$

$$+\text{Coupling}(\text{lambdaH1Inv1}, \{\}, 0)\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd2})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2})\}, \{\}))\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})\}, \{\})\text{Bar}(\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})\}, \{\}))\text{CG}(\text{T3HiggsPortal1CGInv1}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)}), \text{Bar}(\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})), \text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Bar}(\text{Index}(\text{d\$\$2}, \text{T3BSMd2}))\})$$

$$+\text{Coupling}(\text{lambdaH1Inv2}, \{\}, 0)\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd2})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2})\}, \{\}))\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})\}, \{\})\text{Bar}(\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})\}, \{\}))\text{CG}(\text{T3HiggsPortal1CGInv2}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)}), \text{Bar}(\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})), \text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Bar}(\text{Index}(\text{d\$\$2}, \text{T3BSMd2}))\})$$

$$+\text{Coupling}(\text{lambdaT3}, \{\}, 0)\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd2})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2})\}, \{\}))\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})\}, \{\})\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})\}, \{\})\text{CG}(\text{T3MixCG}, \{\text{Bar}(\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})), \text{Bar}(\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})), \text{Bar}(\text{Index}(\text{d\$\$2}, \text{T3BSMd2})), \text{Index}(\text{d\$\$1}, \text{T3BSMd2})\})$$

$$+\text{Bar}(\text{Coupling}(\text{lambdaT3ExtraQ0202001}, \{\}, 0)\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2})\}, \{\})\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd2})\}, \{\})\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})\}, \{\})\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})\}, \{\})\text{CG}(\text{Bar}(\text{T3ExtraCGQQ0202001}), \{\text{Bar}(\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})), \text{Bar}(\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})), \text{Bar}(\text{Index}(\text{d\$\$1}, \text{T3BSMd2})), \text{Bar}(\text{Index}(\text{d\$\$2}, \text{T3BSMd2}))\})$$

$$+\text{Coupling}(\text{lambdaT3ExtraQ0012011}, \{\}, 0)\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$4}, \text{T3BSMd2})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2})\}, \{\}))\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd2})\}, \{\}))\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$3}, \text{T3BSMd2})\}, \{\}))\text{CG}(\text{T3ExtraCGQ0012011}, \{\text{Bar}(\text{Index}(\text{d\$\$1}, \text{T3BSMd2})), \text{Index}(\text{d\$\$4}, \text{T3BSMd2}), \text{Index}(\text{d\$\$2}, \text{T3BSMd2}), \text{Index}(\text{d\$\$3}, \text{T3BSMd2})\})$$

$$+\frac{1}{2}\text{Coupling}(\text{lambdaS1Inv1}, \{\}, 0)\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$3}, \text{T3BSMd2})\}, \{\})\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$4}, \text{T3BSMd2})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2})\}, \{\}))\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd2})\}, \{\}))\text{CG}(\text{T3Scalar1SelfCGInv1}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Bar}(\text{Index}(\text{d\$\$3}, \text{T3BSMd2})), \text{Index}(\text{d\$\$2}, \text{T3BSMd2}), \text{Bar}(\text{Index}(\text{d\$\$4}, \text{T3BSMd2}))\})$$

$$+\text{Coupling}(\text{MS2}, \{\}, 0)^2\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2})\}, \{\})(-\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2})\}, \{\}))$$
$$+\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2})\}, \{\text{Index}(\text{d\$\$1}, \text{Lorentz})\})\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2})\}, \{\text{Index}(\text{d\$\$1}, \text{Lorentz})\}))$$

$$+\text{Bar}(\text{Coupling}(\text{lambdaT3}, \{\}, 0)\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd2})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2})\}, \{\}))\text{Bar}(\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})\}, \{\}))\text{Bar}(\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})\}, \{\}))\text{CG}(\text{Bar}(\text{T3MixCG}), \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)}), \text{Index}(\text{d\$\$2}, \text{SU2L(fund)}), \text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Bar}(\text{Index}(\text{d\$\$2}, \text{T3BSMd2}))\})$$

$$+\text{Coupling}(\text{lambdaH2Inv1}, \{\}, 0)\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd2})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2})\}, \{\}))\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})\}, \{\})\text{Bar}(\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})\}, \{\}))\text{CG}(\text{T3HiggsPortal2CGInv1}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)}), \text{Bar}(\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})), \text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Bar}(\text{Index}(\text{d\$\$2}, \text{T3BSMd2}))\})$$

$$+\text{Coupling}(\text{lambdaH2Inv2}, \{\}, 0)\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd2})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2})\}, \{\}))\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})\}, \{\})\text{Bar}(\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})\}, \{\}))\text{CG}(\text{T3HiggsPortal2CGInv2}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)}), \text{Bar}(\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})), \text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Bar}(\text{Index}(\text{d\$\$2}, \text{T3BSMd2}))\})$$

$$+\text{Bar}(\text{Coupling}(\text{lambdaT3ExtraM000101I1}, \{\}, 0))\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2})\}, \{\})\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd2})\}, \{\})\text{CG}(\text{Bar}(\text{T3ExtraCGM000101I1}), \{\text{Bar}(\text{Index}(\text{d\$\$1}, \text{T3BSMd2})), \text{Bar}(\text{Index}(\text{d\$\$2}, \text{T3BSMd2}))\})$$

$$+\text{Bar}(\text{Coupling}(\text{lambdaT3ExtraQ110101I1}, \{\}, 0)\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2})\}, \{\})\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd2})\}, \{\})\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})\}, \{\})\text{Bar}(\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})\}, \{\}))\text{CG}(\text{Bar}(\text{T3ExtraCGQ110101I1}), \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)}), \text{Bar}(\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})), \text{Bar}(\text{Index}(\text{d\$\$1}, \text{T3BSMd2})), \text{Bar}(\text{Index}(\text{d\$\$2}, \text{T3BSMd2}))\})$$

$$+\text{Bar}(\text{Coupling}(\text{lambdaT3ExtraQ110101I2}, \{\}, 0)\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2})\}, \{\})\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd2})\}, \{\})\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})\}, \{\})\text{Bar}(\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})\}, \{\}))\text{CG}(\text{Bar}(\text{T3ExtraCGQ110101I2}), \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)}), \text{Bar}(\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})), \text{Bar}(\text{Index}(\text{d\$\$1}, \text{T3BSMd2})), \text{Bar}(\text{Index}(\text{d\$\$2}, \text{T3BSMd2}))\})$$

$$+\text{Coupling}(\text{lambdaT3ExtraQ0202001}, \{\}, 0)\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2})\}, \{\})\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd2})\}, \{\})\text{Bar}(\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})\}, \{\}))\text{Bar}(\text{Field}(H, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{SU2L(fund)})\}, \{\}))\text{CG}(\text{T3ExtraCGQQ0202001}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)}), \text{Index}(\text{d\$\$2}, \text{SU2L(fund)}), \text{Bar}(\text{Index}(\text{d\$\$1}, \text{T3BSMd2})), \text{Bar}(\text{Index}(\text{d\$\$2}, \text{T3BSMd2}))\})$$

$$+\text{Coupling}(\text{lambdaT3ExtraQ0012I1}, \{\}, 0)\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$4}, \text{T3BSMd2})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2})\}, \{\}))\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd2})\}, \{\}))\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$3}, \text{T3BSMd2})\}, \{\}))\text{CG}(\text{T3ExtraCGQ0012I1}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Bar}(\text{Index}(\text{d\$\$4}, \text{T3BSMd2})), \text{Index}(\text{d\$\$2}, \text{T3BSMd2}), \text{Index}(\text{d\$\$3}, \text{T3BSMd2})\})$$

$$+\text{Coupling}(\text{lambda12Inv1}, \{\}, 0)\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$3}, \text{T3BSMd2})\}, \{\})\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$4}, \text{T3BSMd2})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2})\}, \{\}))\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd2})\}, \{\}))\text{CG}(\text{T3CrossScalarCGInv1}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Bar}(\text{Index}(\text{d\$\$3}, \text{T3BSMd2})), \text{Index}(\text{d\$\$2}, \text{T3BSMd2}), \text{Bar}(\text{Index}(\text{d\$\$4}, \text{T3BSMd2}))\})$$

$$+\text{Coupling}(\text{lambda12Inv2}, \{\}, 0)\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$3}, \text{T3BSMd2})\}, \{\})\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$4}, \text{T3BSMd2})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2})\}, \{\}))\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd2})\}, \{\}))\text{CG}(\text{T3CrossScalarCGInv2}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Bar}(\text{Index}(\text{d\$\$3}, \text{T3BSMd2})), \text{Index}(\text{d\$\$2}, \text{T3BSMd2}), \text{Bar}(\text{Index}(\text{d\$\$4}, \text{T3BSMd2}))\})$$

$$+\text{Bar}(\text{Coupling}(\text{lambdaT3ExtraQ0012011}, \{\}, 0)\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd2})\}, \{\})\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$3}, \text{T3BSMd2})\}, \{\})\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$4}, \text{T3BSMd2})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2})\}, \{\}))\text{CG}(\text{Bar}(\text{T3ExtraCGQ0012011}), \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Bar}(\text{Index}(\text{d\$\$2}, \text{T3BSMd2})), \text{Bar}(\text{Index}(\text{d\$\$3}, \text{T3BSMd2})), \text{Bar}(\text{Index}(\text{d\$\$4}, \text{T3BSMd2}))\})$$

$$+\frac{1}{2}\text{Coupling}(\text{lambdaS2Inv1}, \{\}, 0)\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$3}, \text{T3BSMd2})\}, \{\})\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$4}, \text{T3BSMd2})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2})\}, \{\}))\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd2})\}, \{\}))\text{CG}(\text{T3Scalar2SelfCGInv1}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Bar}(\text{Index}(\text{d\$\$3}, \text{T3BSMd2})), \text{Index}(\text{d\$\$2}, \text{T3BSMd2}), \text{Bar}(\text{Index}(\text{d\$\$4}, \text{T3BSMd2}))\})$$

$$+\text{Bar}(\text{Coupling}(\text{lambdaT3ExtraQ0012I21}, \{\}, 0)\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd2})\}, \{\})\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$3}, \text{T3BSMd2})\}, \{\})\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$4}, \text{T3BSMd2})\}, \{\})\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2})\}, \{\}))\text{CG}(\text{Bar}(\text{T3ExtraCGQ0012I21}), \{\text{Bar}(\text{Index}(\text{d\$\$2}, \text{T3BSMd2})), \text{Index}(\text{d\$\$1}, \text{T3BSMd2}), \text{Bar}(\text{Index}(\text{d\$\$3}, \text{T3BSMd2})), \text{Bar}(\text{Index}(\text{d\$\$4}, \text{T3BSMd2}))\})$$

$$+\text{Bar}(\text{Coupling}(\text{lambdaT3ExtraQ0002021}, \{\}, 0)\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2})\}, \{\})\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$2}, \text{T3BSMd2})\}, \{\})\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$3}, \text{T3BSMd2})\}, \{\})\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$4}, \text{T3BSMd2})\}, \{\})\text{CG}(\text{Bar}(\text{T3ExtraCGQ0002021}), \{\text{Bar}(\text{Index}(\text{d\$\$1}, \text{T3BSMd2})), \text{Bar}(\text{Index}(\text{d\$\$2}, \text{T3BSMd2})), \text{Bar}(\text{Index}(\text{d\$\$3}, \text{T3BSMd2})), \text{Bar}(\text{Index}(\text{d\$\$4}, \text{T3BSMd2}))\})$$

$$+\text{Coupling}(\text{y2}, \{\text{Index}(\text{d\$\$1}, \text{Flavor}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, 0)\text{Bar}(\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2})\}, \{\}))\text{CG}(\text{T3Y2CG}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)}), \text{Index}(\text{d\$\$1}, \text{T3BSMd3}), \text{Index}(\text{d\$\$1}, \text{T3BSMd2})\})\text{NCM}(\text{Bar}(\text{Field}(l, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})\}, \text{Index}(\text{d\$\$1}, \text{Flavor})\}, \{\})), \text{DiracProduct}(\text{Proj}(1)), \text{Field}(\text{NewFermion}, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, \{\}))$$

$$+\text{Coupling}(\text{y1}, \{\text{Index}(\text{d\$\$1}, \text{Flavor}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, 0)\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2})\}, \{\})\text{CG}(\text{T3Y1CG}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)}), \text{Index}(\text{d\$\$1}, \text{T3BSMd3}), \text{Bar}(\text{Index}(\text{d\$\$1}, \text{T3BSMd2}))\})\text{NCM}(\text{Bar}(\text{Field}(l, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})\}, \text{Index}(\text{d\$\$1}, \text{Flavor})\}, \{\})), \text{DiracProduct}(\text{Proj}(1)), \text{Field}(\text{NewFermion}, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, \{\}))$$

$$-\frac{1}{2}\text{Coupling}(\text{MF}, \{\}, 0)\text{NCM}(\text{Transp}(\text{Field}(\text{NewFermion}, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, \{\})), \text{DiracProduct}(\text{GammaCC}), \text{Field}(\text{NewFermion}, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, \{\}))$$

$$+\frac{1}{2}i\text{NCM}(\text{Transp}(\text{Field}(\text{NewFermion}, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, \{\})), \text{DiracProduct}(\text{GammaCC}, \text{GammaM}(\text{Index}(\text{d\$\$1}, \text{Lorentz}))), \text{Field}(\text{NewFermion}, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, \{\text{Index}(\text{d\$\$1}, \text{Lorentz})\}))$$

$$+\text{Bar}(\text{Coupling}(\text{y1}, \{\text{Index}(\text{d\$\$1}, \text{Flavor}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, 0)\text{Bar}(\text{Field}(\text{NewScalar1}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2})\}, \{\}))\text{CG}(\text{Bar}(\text{T3Y1CG}), \{\text{Bar}(\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})), \text{Index}(\text{d\$\$1}, \text{T3BSMd3}), \text{Index}(\text{d\$\$1}, \text{T3BSMd2})\})\text{NCM}(\text{Transp}(\text{Field}(\text{NewFermion}, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, \{\})), \text{DiracProduct}(\text{GammaCC}, \text{Proj}(-1)), \text{Field}(l, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})\}, \text{Index}(\text{d\$\$1}, \text{Flavor})\}, \{\}))$$

$$+\text{Field}(\text{NewScalar2}, \text{Scalar}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd2})\}, \{\})\text{Bar}(\text{Coupling}(\text{y2}, \{\text{Index}(\text{d\$\$1}, \text{Flavor}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, 0)\text{CG}(\text{Bar}(\text{T3Y2CG}), \{\text{Bar}(\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})), \text{Index}(\text{d\$\$1}, \text{T3BSMd3}), \text{Bar}(\text{Index}(\text{d\$\$1}, \text{T3BSMd2}))\})\text{NCM}(\text{Transp}(\text{Field}(\text{NewFermion}, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{T3BSMd3}), \text{Index}(\text{d\$\$1}, \text{NFlavor})\}, \{\})), \text{DiracProduct}(\text{GammaCC}, \text{Proj}(-1)), \text{Field}(l, \text{Fermion}, \{\text{Index}(\text{d\$\$1}, \text{SU2L(fund)})\}, \text{Index}(\text{d\$\$1}, \text{Flavor})\}, \{\}))$$

Interaction inventory

- `Yukawa1_GenericCG`
- `Yukawa2_GenericCG`
- `Scalar1SelfInv1`
- `Scalar2SelfInv1`
- `HiggsPortal1Inv1`
- `HiggsPortal1Inv2`
- `HiggsPortal2Inv1`
- `HiggsPortal2Inv2`
- `ScalarCrossPortalInv1`

- ScalarCrossPortalInv2
- T3ScalarMix_GenericCG
- T3ExtraQ_0_0_0_1_1_2_1
- T3ExtraQ_0_0_0_2_0_2_1
- T3ExtraQ_0_0_1_2_0_1_1
- T3ExtraQ_0_2_0_0_2_0_1
- T3ExtraQ_0_2_0_2_0_0_1
- T3ExtraQ_1_1_0_1_0_1_1
- T3ExtraQ_1_1_0_1_0_1_2
- T3ExtraM_0_0_0_1_0_1_1

One-loop renormalisable beta functions

$16\pi^2\, dX/d\ln\mu = \beta_X$

g_1

$$16\pi^2 \frac{dg_1}{d\ln\mu} = \frac{43{g_1}^3}{6}$$

g_2

$$16\pi^2 \frac{dg_2}{d\ln\mu} = \frac{7{g_2}^3}{6}$$

g_3

$$16\pi^2 \frac{dg_3}{d\ln\mu} = -7{g_3}^3$$

Y_u

$$\begin{aligned}
16\pi^2 \frac{dY_u}{d\ln\mu} = & \, 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_u)_{ij} \\
& + \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_u)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_u)_{ij} \\
& - \frac{3}{2}(\Big(Y_d\ (Y_d)^\dagger\ Y_u\Big)_{ij} - \Big(Y_u\ (Y_u)^\dagger\ Y_u\Big)_{ij}) \\
& - \frac{9}{4}{g_2}^2\,(Y_u)_{ij} \\
& - 8{g_3}^2\,(Y_u)_{ij} \\
& - \frac{17}{12}{g_1}^2\,(Y_u)_{ij}
\end{aligned}$$

Y_d

$$\begin{aligned}
16\pi^2 \frac{dY_d}{d\ln\mu} = & \, \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij} \\
& + \frac{3}{2}(\Big(Y_d\ (Y_d)^\dagger\ Y_d\Big)_{ij} - \Big(Y_u\ (Y_u)^\dagger\ Y_d\Big)_{ij}) \\
& - \frac{9}{4}{g_2}^2\,(Y_d)_{ij} \\
& - 8{g_3}^2\,(Y_d)_{ij} \\
& - \frac{5}{12}{g_1}^2\,(Y_d)_{ij}
\end{aligned}$$

Y_e

$$\begin{aligned}16\pi^2 \frac{dY_e}{d\ln\mu} &= 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_e)_{ij} \\ &+ 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_e)_{ij} \\ &+ \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_e)_{ij} \\ &+ \frac{3}{4}(\Big(y_1\ (y_1)^\dagger\ Y_e\Big)_{ij} + \Big(y_2\ (y_2)^\dagger\ Y_e\Big)_{ij} + 2\,\Big(Y_e\ (Y_e)^\dagger\ Y_e\Big)_{ij}) \\ &- \frac{9}{4}{g_2}^2\,(Y_e)_{ij} \\ &- \frac{15}{4}{g_1}^2\,(Y_e)_{ij}\end{aligned}$$

y_1

$$\begin{aligned}16\pi^2 \frac{dy_1}{d\ln\mu} &= \frac{3}{2}\,\mathrm{Tr}\Big(y_1\ (y_2)^\dagger\Big)\,(y_2)_{ij} \\ &+ \frac{3}{2}T_\nu^{(1)}\,(y_1)_{ij} \\ &+ \frac{1}{4}(2\,\Big(y_1\ (y_2)^\dagger\ y_2\Big)_{ij} + 3\,\Big(y_2\ (y_2)^\dagger\ y_1\Big)_{ij} + 2\,\Big(Y_e\ (Y_e)^\dagger\ y_1\Big)_{ij} + 5\,\Big(y_1\ (y_1)^\dagger\ y_1\Big)_{ij}) \\ &- \frac{33}{4}{g_2}^2\,(y_1)_{ij} \\ &- \frac{3}{4}{g_1}^2\,(y_1)_{ij}\end{aligned}$$

y_2

$$\begin{aligned}16\pi^2 \frac{dy_2}{d\ln\mu} &= \frac{3}{2}\,\mathrm{Tr}\Big(y_2\ (y_1)^\dagger\Big)\,(y_1)_{ij} \\ &+ \frac{3}{2}T_\nu^{(2)}\,(y_2)_{ij} \\ &+ \frac{1}{4}(3\,\Big(y_1\ (y_1)^\dagger\ y_2\Big)_{ij} + 2\,\Big(y_2\ (y_1)^\dagger\ y_1\Big)_{ij} + 2\,\Big(Y_e\ (Y_e)^\dagger\ y_2\Big)_{ij} + 5\,\Big(y_2\ (y_2)^\dagger\ y_2\Big)_{ij}) \\ &- \frac{33}{4}{g_2}^2\,(y_2)_{ij} \\ &- \frac{3}{4}{g_1}^2\,(y_2)_{ij}\end{aligned}$$

M_F

$$\begin{aligned}16\pi^2 \frac{dM_F}{d\ln\mu} &= \frac{1}{2}(\Big(\left(M_F\right)^T\ (y_1)^\dagger\ y_1\Big)_{ij} + \Big(\left(y_1\right)^T\ (y_1)^*\ M_F\Big)_{ij} + \Big(\left(M_F\right)^T\ (y_2)^\dagger\ y_2\Big)_{ij} + \Big(\left(y_2\right)^T\ (y_2)^*\ M_F\Big)_{ij}) \\ &- 6{g_2}^2(\Big(\left(M_F\right)^T\Big)_{ij} + (M_F)_{ij})\end{aligned}$$

m_1^2

$$\begin{aligned}16\pi^2 \frac{dm_1^2}{d\ln\mu} &= m_1^2(3T_\nu^{(1)} + 6\lambda_{S_1}^{(1)}) \\ &- 12\,\mathrm{Tr}\Big(M_F\ (M_F)^*\ \left(y_1\right)^T\ (y_1)^*\Big)\end{aligned}$$

$$
\begin{aligned}
&- 12\,\mathrm{Tr}\Big(M_F\ (y_1)^\dagger\ y_1\ (M_F)^*\Big)\\
&+ 6\mathrm{muT3GeneralS1S2}\,(\lambda_{11})^*\\
&+ 6\lambda_{11}\,(\mathrm{muT3GeneralS1S2})^*\\
&- \frac{9{g_2}^2 m_1^2}{2}\\
&- \frac{3{g_1}^2 m_1^2}{2}\\
&+ 4\lambda_{12}^{(1)} m_2^2
\end{aligned}
$$

m_2^2

$$
\begin{aligned}
16\pi^2\,\frac{dm_2^2}{d\ln\mu} = {} & m_2^2(3T_\nu^{(2)} + 6\lambda_{S_2}^{(1)})\\
&- 12\,\mathrm{Tr}\Big(M_F\ (M_F)^*\ (y_2)^T\ (y_2)^*\Big)\\
&- 12\,\mathrm{Tr}\Big(M_F\ (y_2)^\dagger\ y_2\ (M_F)^*\Big)\\
&+ 6\mathrm{muT3GeneralS1S2}\,(\lambda_{10})^*\\
&+ 6\lambda_{10}\,(\mathrm{muT3GeneralS1S2})^*\\
&- \frac{9{g_2}^2 m_2^2}{2}\\
&- \frac{3{g_1}^2 m_2^2}{2}\\
&+ 4\lambda_{12}^{(1)} m_1^2
\end{aligned}
$$

λ_1

$$
\begin{aligned}
16\pi^2\,\frac{d\lambda_1}{d\ln\mu} = {} & \lambda_1(12\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 4\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 12\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big))\\
&- 12\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\Big)\\
&- 4\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\Big)\\
&- 12\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\Big)\\
&+ 8\lambda_7\,(\lambda_7)^*\\
&+ 4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}\right)^*\\
&+ 8\lambda_8\,(\lambda_8)^*\\
&+ 4\lambda_5\,(\lambda_5)^*\\
&+ 4\mathrm{lambdaT3GeneralQ110101I1}\,(\mathrm{lambdaT3GeneralQ110101I1})^*\\
&+ \frac{9{g_2}^4}{4}\\
&+ {g_1}^2\left(\frac{3{g_2}^2}{2} - 3\lambda_1\right)\\
&- 9{g_2}^2\lambda_1\\
&+ \frac{3{g_1}^4}{4}\\
&+ 12{\lambda_1}^2\\
&+ 4{\lambda_{HS_1}^{(1)}}^2\\
&+ \frac{{\lambda_{HS_1}^{(A)}}^2}{4}
\end{aligned}
$$

$$+ 4{\lambda_{HS_2}^{(1)}}^2$$
$$+ \frac{{\lambda_{HS_2}^{(A)}}^2}{4}$$

$\lambda_{S_1}^{(1)}$

$$\begin{aligned}
16\pi^2 \frac{d\lambda_{S_1}^{(1)}}{d\ln\mu} = & 6\lambda_{S_1}^{(1)} T_\nu^{(1)} \\
& - 5\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_1\ (y_1)^\dagger\Big) \\
& + 8\lambda_7\,(\lambda_7)^* \\
& + 8\lambda_9\,(\lambda_9)^* \\
& + 24\lambda_{11}\,(\lambda_{11})^* \\
& + \frac{9{g_2}^4}{4} \\
& + {g_1}^2\left(\frac{3{g_2}^2}{2} - 3\lambda_{S_1}^{(1)}\right) \\
& - 9{g_2}^2\lambda_{S_1}^{(1)} \\
& + \frac{3{g_1}^4}{4} \\
& + 4{\lambda_{12}^{(1)}}^2 \\
& + \frac{{\lambda_{12}^{(A)}}^2}{4} \\
& + 4{\lambda_{HS_1}^{(1)}}^2 \\
& + \frac{{\lambda_{HS_1}^{(A)}}^2}{4} \\
& + 12{\lambda_{S_1}^{(1)}}^2
\end{aligned}$$

$\lambda_{S_2}^{(1)}$

$$\begin{aligned}
16\pi^2 \frac{d\lambda_{S_2}^{(1)}}{d\ln\mu} = & 6\lambda_{S_2}^{(1)} T_\nu^{(2)} \\
& - 5\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ y_2\ (y_2)^\dagger\Big) \\
& + 8\lambda_8\,(\lambda_8)^* \\
& + 8\lambda_9\,(\lambda_9)^* \\
& + 24\lambda_{10}\,(\lambda_{10})^* \\
& + \frac{9{g_2}^4}{4} \\
& + {g_1}^2\left(\frac{3{g_2}^2}{2} - 3\lambda_{S_2}^{(1)}\right) \\
& - 9{g_2}^2\lambda_{S_2}^{(1)} \\
& + \frac{3{g_1}^4}{4} \\
& + 4{\lambda_{12}^{(1)}}^2 \\
& + \frac{{\lambda_{12}^{(A)}}^2}{4}
\end{aligned}$$

$$
\begin{aligned}
&+ 4{\lambda_{HS_2}^{(1)}}^2 \\
&+ \frac{{\lambda_{HS_2}^{(A)}}^2}{4} \\
&+ 12{\lambda_{S_2}^{(1)}}^2
\end{aligned}
$$

$\lambda_{HS_1}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{HS_1}^{(1)}}{d\ln\mu} = {} & \lambda_{HS_1}^{(1)}(3T_\nu^{(1)} + 6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)) \\
& - 3\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\Big) \\
& + 24\lambda_7\,(\lambda_7)^* \\
& + 3\lambda_{11}(\Big(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}\Big)^* - (\mathrm{lambdaT3GeneralQ110101I1})^*) \\
& + \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}(4\,\Big(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}\Big)^* + 3\,(\lambda_{11})^* + 2\,(\mathrm{lambdaT3GeneralQ110101I1})^*) \\
& + \mathrm{lambdaT3GeneralQ110101I1}(2\,\Big(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}\Big)^* - 3\,(\lambda_{11})^* + 4\,(\mathrm{lambdaT3GeneralQ110101I1})^*) \\
& + 6\lambda_5\,(\lambda_5)^* \\
& + \frac{9{g_2}^4}{4} \\
& - 9{g_2}^2\lambda_{HS_1}^{(1)} \\
& + \frac{3{g_1}^4}{4} \\
& - 3{g_1}^2\lambda_{HS_1}^{(1)} \\
& + 4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)} \\
& + 6\lambda_1\lambda_{HS_1}^{(1)} \\
& + 4{\lambda_{HS_1}^{(1)}}^2 \\
& + 6\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)} \\
& + \frac{3{\lambda_{HS_1}^{(A)}}^2}{4}
\end{aligned}
$$

$\lambda_{HS_2}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{HS_2}^{(1)}}{d\ln\mu} = {} & \lambda_{HS_2}^{(1)}(3T_\nu^{(2)} + 6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)) \\
& - 3\,\mathrm{Tr}\Big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\Big) \\
& + 3\lambda_{10}(\Big(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}\Big)^* - (\mathrm{lambdaT3GeneralQ110101I1})^*) \\
& + \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}(4\,\Big(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}\Big)^* + 3\,(\lambda_{10})^* + 2\,(\mathrm{lambdaT3GeneralQ110101I1})^*) \\
& + \mathrm{lambdaT3GeneralQ110101I1}(2\,\Big(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}\Big)^* - 3\,(\lambda_{10})^* + 4\,(\mathrm{lambdaT3GeneralQ110101I1})^*) \\
& + 24\lambda_8\,(\lambda_8)^* \\
& + 6\lambda_5\,(\lambda_5)^* \\
& + \frac{9{g_2}^4}{4} \\
& - 9{g_2}^2\lambda_{HS_2}^{(1)}
\end{aligned}
$$

$$
\begin{aligned}
&+ \frac{3{g_1}^4}{4} \\
&- 3{g_1}^2\lambda_{HS_2}^{(1)} \\
&+ 4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)} \\
&+ 6\lambda_1\lambda_{HS_2}^{(1)} \\
&+ 4{\lambda_{HS_2}^{(1)}}^2 \\
&+ 6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)} \\
&+ \frac{3{\lambda_{HS_2}^{(A)}}^2}{4}
\end{aligned}
$$

$\lambda_{12}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{12}^{(1)}}{d\ln\mu} = {} & \lambda_{12}^{(1)}(3T_\nu^{(1)} + 3T_\nu^{(2)}) \\
& - \frac{9}{2}\operatorname{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big) \\
& - 3\operatorname{Tr}\Big(y_1\ (y_2)^\dagger\ y_2\ (y_1)^\dagger\Big) \\
& + \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^* + (\text{lambdaT3GeneralQ110101I1})^*) \\
& + \text{lambdaT3GeneralQ110101I1}(\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^* + 2\,(\text{lambdaT3GeneralQ110101I1})^*) \\
& + 24\lambda_9\,(\lambda_9)^* \\
& + 9\lambda_{10}((\lambda_{10})^* + (\lambda_{11})^*) \\
& + 9\lambda_{11}((\lambda_{10})^* + (\lambda_{11})^*) \\
& + 3\lambda_5\,(\lambda_5)^* \\
& + \frac{9{g_2}^4}{4} \\
& - 9{g_2}^2\lambda_{12}^{(1)} \\
& + \frac{3{g_1}^4}{4} \\
& - 3{g_1}^2\lambda_{12}^{(1)} \\
& + 4{\lambda_{12}^{(1)}}^2 \\
& + 6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)} \\
& + 6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)} \\
& + \frac{3{\lambda_{12}^{(A)}}^2}{4} \\
& + 4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}
\end{aligned}
$$

λ_5

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_5}{d\ln\mu} = {} & \lambda_5\left(\frac{3}{2}T_\nu^{(1)} + \frac{3}{2}T_\nu^{(2)} + 6\operatorname{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\operatorname{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\operatorname{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) + \frac{\lambda_{12}^{(A)}}{6} + \frac{2\lambda_{HS_1}^{(A)}}{3} + 2\lambda_{HS_2}^{(A)}\right) \\
& - (\lambda_7)^*\operatorname{Tr}\Big(y_2\ (y_1)^\dagger\Big) \\
& + 3\,(\lambda_8)^*\operatorname{Tr}\Big(y_1\ (y_2)^\dagger\Big) \\
& - 9{g_2}^2\lambda_5
\end{aligned}
$$

$$
\begin{aligned}
&- 3{g_1}^2\lambda_5 \\
&+ 2\lambda_{12}^{(1)}\lambda_5 \\
&+ 2\lambda_1\lambda_5 \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_5 \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_5
\end{aligned}
$$

$\lambda_{HS_1}^{(A)}$

$$
\begin{aligned}
16\pi^2\,\frac{d\lambda_{HS_1}^{(A)}}{d\ln\mu} = {}& \lambda_{HS_1}^{(A)}(3T_\nu^{(1)} + 6\,\mathrm{Tr}\Big(Y_d\ {(Y_d)}^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ {(Y_e)}^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ {(Y_u)}^\dagger\Big)) \\
&- \frac{4}{3}\,\mathrm{Tr}\Big(y_1\ {(y_1)}^\dagger\ Y_e\ {(Y_e)}^\dagger\Big) \\
&- 64\lambda_7\,{(\lambda_7)}^* \\
&+ 4\lambda_{11}(\Big(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}\Big)^* + (\mathrm{lambdaT3GeneralQ110101I1})^*) \\
&+ 4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}(2\,\Big(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}\Big)^* + {(\lambda_{11})}^*) \\
&+ 4\mathrm{lambdaT3GeneralQ110101I1}((\lambda_{11})^* - 2\,(\mathrm{lambdaT3GeneralQ110101I1})^*) \\
&+ \frac{16}{3}\lambda_5\,{(\lambda_5)}^* \\
&+ {g_1}^2\,\Big(-6{g_2}^2 - 3\lambda_{HS_1}^{(A)}\Big) \\
&- 9{g_2}^2\lambda_{HS_1}^{(A)} \\
&+ \lambda_{12}^{(A)}\lambda_{HS_2}^{(A)} \\
&+ 2\lambda_1\lambda_{HS_1}^{(A)} \\
&+ 8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)} \\
&+ 2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}
\end{aligned}
$$

$\lambda_{HS_2}^{(A)}$

$$
\begin{aligned}
16\pi^2\,\frac{d\lambda_{HS_2}^{(A)}}{d\ln\mu} = {}& \lambda_{HS_2}^{(A)}(3T_\nu^{(2)} + 6\,\mathrm{Tr}\Big(Y_d\ {(Y_d)}^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ {(Y_e)}^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ {(Y_u)}^\dagger\Big)) \\
&- 4\,\mathrm{Tr}\Big(y_2\ {(y_2)}^\dagger\ Y_e\ {(Y_e)}^\dagger\Big) \\
&- 4\lambda_{10}(\Big(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}\Big)^* + (\mathrm{lambdaT3GeneralQ110101I1})^*) \\
&- 4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}(2\,\Big(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}\Big)^* + {(\lambda_{10})}^*) \\
&+ 64\lambda_8\,{(\lambda_8)}^* \\
&+ \mathrm{lambdaT3GeneralQ110101I1}(8\,(\mathrm{lambdaT3GeneralQ110101I1})^* - 4\,{(\lambda_{10})}^*) \\
&+ 16\lambda_5\,{(\lambda_5)}^* \\
&+ {g_1}^2\,\Big(6{g_2}^2 - 3\lambda_{HS_2}^{(A)}\Big) \\
&- 9{g_2}^2\lambda_{HS_2}^{(A)} \\
&+ \lambda_{12}^{(A)}\lambda_{HS_1}^{(A)} \\
&+ 2\lambda_1\lambda_{HS_2}^{(A)} \\
&+ 8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)} \\
&+ 2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}
\end{aligned}
$$

$\lambda_{12}^{(A)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{12}^{(A)}}{d\ln\mu} = \lambda_{12}^{(A)}(3T_\nu^{(1)} + 3T_\nu^{(2)}) \\
&- \frac{2}{3}\,\mathrm{Tr}\Big(y_1\ (y_1)^\dagger\ y_2\ (y_2)^\dagger\Big) \\
&- \frac{8}{3}\,\mathrm{Tr}\Big(y_1\ (y_2)^\dagger\ y_2\ (y_1)^\dagger\Big) \\
&+ 4\mathrm{lambdaT3GeneralQ110101I1}\left(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* \\
&+ 4\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\,(\mathrm{lambdaT3GeneralQ110101I1})^* \\
&- 64\lambda_9\,(\lambda_9)^* \\
&- 4\lambda_{10}(5\,(\lambda_{10})^* + (\lambda_{11})^*) \\
&- 4\lambda_{11}((\lambda_{10})^* + 5\,(\lambda_{11})^*) \\
&+ \frac{4}{3}\lambda_5\,(\lambda_5)^* \\
&+ {g_1}^2\left(-6{g_2}^2 - 3\lambda_{12}^{(A)}\right) \\
&- 9{g_2}^2\lambda_{12}^{(A)} \\
&+ 8\lambda_{12}^{(1)}\lambda_{12}^{(A)} \\
&+ 2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)} \\
&+ 2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)} \\
&+ \lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}
\end{aligned}
$$

λ_8

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_8}{d\ln\mu} = \lambda_8(3T_\nu^{(2)} + 6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)) \\
&+ \frac{3}{2}\,(\lambda_5)^*\,\mathrm{Tr}\Big(y_1\ (y_2)^\dagger\Big) \\
&+ 4\lambda_7\,(\lambda_9)^* \\
&- 9{g_2}^2\lambda_8 \\
&- 3{g_1}^2\lambda_8 \\
&+ 2\lambda_1\lambda_8 \\
&+ 8\lambda_{HS_2}^{(1)}\lambda_8 \\
&+ 4\lambda_{HS_2}^{(A)}\lambda_8 \\
&+ 2\lambda_8\lambda_{S_2}^{(1)}
\end{aligned}
$$

λ_7

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_7}{d\ln\mu} = \lambda_7(3T_\nu^{(1)} + 6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)) \\
&- \frac{1}{2}\,(\lambda_5)^*\,\mathrm{Tr}\Big(y_2\ (y_1)^\dagger\Big) \\
&- 9{g_2}^2\lambda_7 \\
&- 3{g_1}^2\lambda_7 \\
&+ 2\lambda_1\lambda_7
\end{aligned}
$$

$$
\begin{aligned}
&+ 8\lambda_{HS_1}^{(1)}\lambda_7 \\
&- 4\lambda_{HS_1}^{(A)}\lambda_7 \\
&+ 2\lambda_7\lambda_{S_1}^{(1)} \\
&+ 4\lambda_8\lambda_9
\end{aligned}
$$

λ_9

$$
\begin{aligned}
16\pi^2\,\frac{d\lambda_9}{d\ln\mu} = {} & \lambda_9(3T_\nu^{(1)} + 3T_\nu^{(2)}) \\
& + \frac{5}{6}\,\mathrm{Tr}\Big(y_2\ (y_1)^\dagger\ y_2\ (y_1)^\dagger\Big) \\
& + 4\lambda_7\,(\lambda_8)^* \\
& - 9{g_2}^2\lambda_9 \\
& - 3{g_1}^2\lambda_9 \\
& + 8\lambda_{12}^{(1)}\lambda_9 \\
& - 4\lambda_{12}^{(A)}\lambda_9 \\
& + {\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}}^2 \\
& + 2\lambda_{S_1}^{(1)}\lambda_9 \\
& + 2\lambda_9\lambda_{S_2}^{(1)} \\
& + 5{\lambda_{10}}^2 \\
& + 2\lambda_{10}\lambda_{11} \\
& + 5{\lambda_{11}}^2 \\
& + \mathrm{lambdaT3GeneralQ110101I1}^2
\end{aligned}
$$

λ_{10}

$$
\begin{aligned}
16\pi^2\,\frac{d\lambda_{10}}{d\ln\mu} = {} & \lambda_{10}\left(\frac{3}{2}T_\nu^{(1)} + \frac{9}{2}T_\nu^{(2)}\right) \\
& + 4\lambda_9(5\,(\lambda_{10})^* + (\lambda_{11})^*) \\
& - 9{g_2}^2\lambda_{10} \\
& - 3{g_1}^2\lambda_{10} \\
& + \lambda_{12}^{(1)}(6\lambda_{10} + 6\lambda_{11}) \\
& + \lambda_{12}^{(A)}\left(-\frac{5\lambda_{10}}{2} - \frac{\lambda_{11}}{2}\right) \\
& + \lambda_{HS_2}^{(1)}(2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}} - 2\mathrm{lambdaT3GeneralQ110101I1}) \\
& + \lambda_{HS_2}^{(A)}\left(-\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}}{2} - \frac{\mathrm{lambdaT3GeneralQ110101I1}}{2}\right) \\
& + 12\lambda_{10}\lambda_{S_2}^{(1)}
\end{aligned}
$$

λ_{11}

$$
\begin{aligned}
16\pi^2\,\frac{d\lambda_{11}}{d\ln\mu} = {} & \lambda_{11}\left(\frac{9}{2}T_\nu^{(1)} + \frac{3}{2}T_\nu^{(2)}\right) \\
& + 4\lambda_9((\lambda_{10})^* + 5\,(\lambda_{11})^*)
\end{aligned}
$$

$$
\begin{aligned}
&- 9{g_2}^2\lambda_{11} \\
&- 3{g_1}^2\lambda_{11} \\
&+ \lambda_{12}^{(1)}(6\lambda_{10} + 6\lambda_{11}) \\
&+ \lambda_{12}^{(A)}\left(-\frac{\lambda_{10}}{2} - \frac{5\lambda_{11}}{2}\right) \\
&+ \lambda_{HS_1}^{(1)}(2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} - 2\text{lambdaT3GeneralQ110101I1}) \\
&+ \lambda_{HS_1}^{(A)}\left(\frac{\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2} + \frac{\text{lambdaT3GeneralQ110101I1}}{2}\right) \\
&+ 12\lambda_{S_1}^{(1)}\lambda_{11}
\end{aligned}
$$

λ_{12}

$$
\begin{aligned}
16\pi^2\,\frac{d\lambda_{12}}{d\ln\mu} = {}& \lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\left(\frac{3}{2}T_\nu^{(1)} + \frac{3}{2}T_\nu^{(2)} + 6\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\right) \\
&+ \lambda_9(8\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^* - 4\,(\text{lambdaT3GeneralQ110101I1})^*) \\
&- 9{g_2}^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} \\
&- 3{g_1}^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} \\
&+ 2\lambda_{12}^{(1)}\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} \\
&+ \lambda_{12}^{(A)}\left(\text{lambdaT3GeneralQ110101I1} - \frac{\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2}\right) \\
&+ \lambda_1(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} - 2\text{lambdaT3GeneralQ110101I1}) \\
&+ \lambda_{HS_1}^{(1)}(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} + 6\lambda_{11}) \\
&+ \lambda_{HS_1}^{(A)}\left(2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} + \frac{\lambda_{11}}{2} + \text{lambdaT3GeneralQ110101I1}\right) \\
&+ \lambda_{HS_2}^{(1)}(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} + 6\lambda_{10}) \\
&+ \lambda_{HS_2}^{(A)}\left(-2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} - \frac{\lambda_{10}}{2} - \text{lambdaT3GeneralQ110101I1}\right)
\end{aligned}
$$

$lambdaT3GeneralQ110101I1$

$$
\begin{aligned}
16\pi^2\,\frac{d\,\text{lambdaT3GeneralQ110101I1}}{d\ln\mu} = {}& \text{lambdaT3GeneralQ110101I1}\left(\frac{3}{2}T_\nu^{(1)} + \frac{3}{2}T_\nu^{(2)} + 6\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\right. \\
&+ \lambda_9(8\,(\text{lambdaT3GeneralQ110101I1})^* - 4\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}\right)^*) \\
&- 9{g_2}^2\text{lambdaT3GeneralQ110101I1} \\
&- 3{g_1}^2\text{lambdaT3GeneralQ110101I1} \\
&+ 2\lambda_{12}^{(1)}\text{lambdaT3GeneralQ110101I1} \\
&+ \lambda_{12}^{(A)}\left(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} - \frac{\text{lambdaT3GeneralQ110101I1}}{2}\right) \\
&+ \lambda_1(4\text{lambdaT3GeneralQ110101I1} - 2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}) \\
&+ \lambda_{HS_1}^{(1)}(4\text{lambdaT3GeneralQ110101I1} - 6\lambda_{11}) \\
&+ \lambda_{HS_1}^{(A)}\left(-\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger} + \frac{\lambda_{11}}{2} - 2\text{lambdaT3GeneralQ110101I1}\right)
\end{aligned}
$$

$$
\begin{aligned}
&+ \lambda_{HS_2}^{(1)}(4\text{lambdaT3GeneralQ110101I1} - 6\lambda_{10}) \\
&+ \lambda_{HS_2}^{(A)}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} - \frac{\lambda_{10}}{2} + 2\text{lambdaT3GeneralQ110101I1}\right)
\end{aligned}
$$

$muT3GeneralS1S2$

$$
\begin{aligned}
16\pi^2 \frac{dmuT3GeneralS1S2}{d\ln\mu} = {} & \text{muT3GeneralS1S2}\left(\frac{3}{2}T_\nu^{(1)} + \frac{3}{2}T_\nu^{(2)}\right) \\
& + 12\lambda_9 \left(\text{muT3GeneralS1S2}\right)^* \\
& - \frac{9{g_2}^2\text{muT3GeneralS1S2}}{2} \\
& - \frac{3{g_1}^2\text{muT3GeneralS1S2}}{2} \\
& + 2\lambda_{12}^{(1)}\text{muT3GeneralS1S2} \\
& - \frac{3\lambda_{12}^{(A)}\text{muT3GeneralS1S2}}{2} \\
& + 6\lambda_{10}m_2^2 \\
& + 6\lambda_{11}m_1^2
\end{aligned}
$$